



INTRODUCTION TO SOFTWARE ENGINEERING

Part I – Overview of Software Engineering

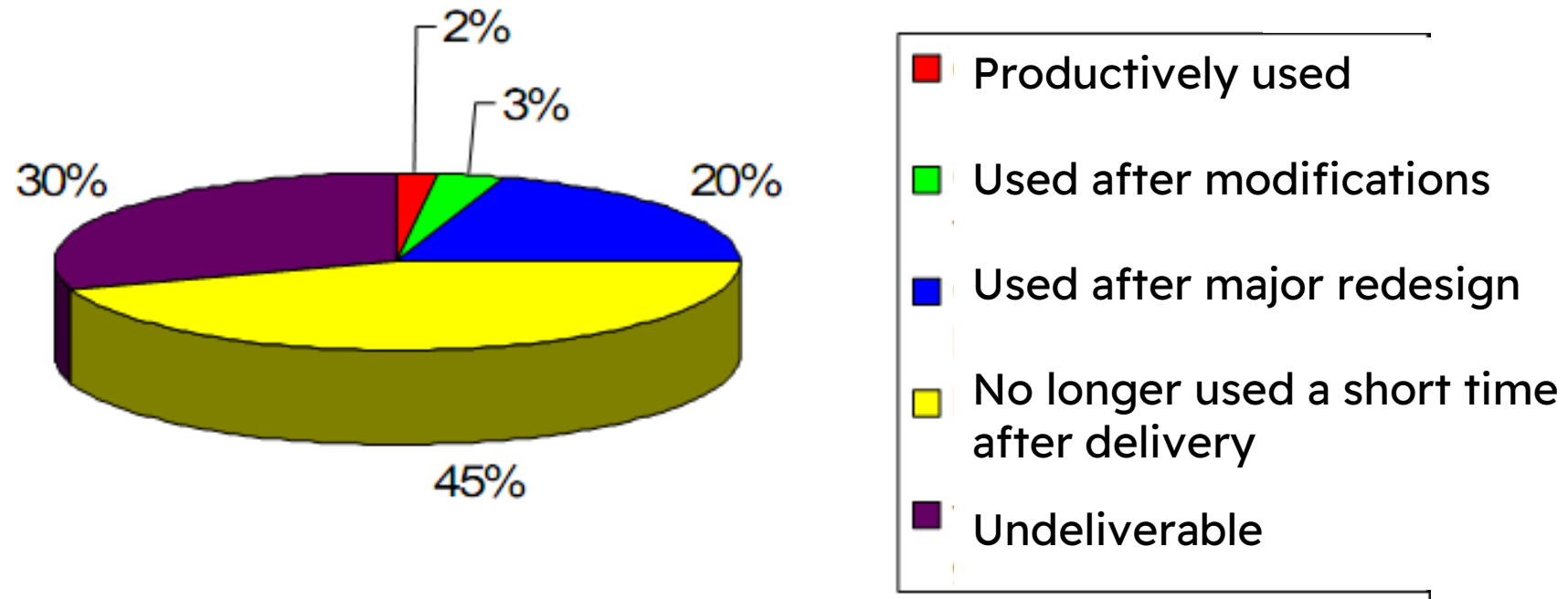
Software Engineering Department
College of Information and Communication Technology

Content

- Why study software engineering?
 - Software crisis
 - Software project success rate
 - Examples of software defects leading to failures
 - Causes of software failures
 - How has software engineering changed?
- What is software engineering?
- Steps of software development
- Who does software engineering?
 - Members of the development team

Why study software engineering?

- **Software crisis** (of the 1960s, 1970s, and 1980s)
 - During the 1970s,



Why study software engineering?

■ Software project success rate

Table I			
Standish project benchmarks over the years			
Year	Successful (%)	Challenged (%)	Failed (%)
1994	16	53	31
1996	27	33	40
1998	26	46	28
2000	28	49	23
2004	29	53	18
2006	35	46	19
2009	32	44	24

Standish Group, CHAOS Report, 2010

- **Successful projects** are on time, on budget, and have a satisfactory implementation.
- **Challenged projects** are over budget, late, and/or have an unsatisfactory implementation.
- **Failed projects** are projects that were either canceled prior to completion or not used after implementation.

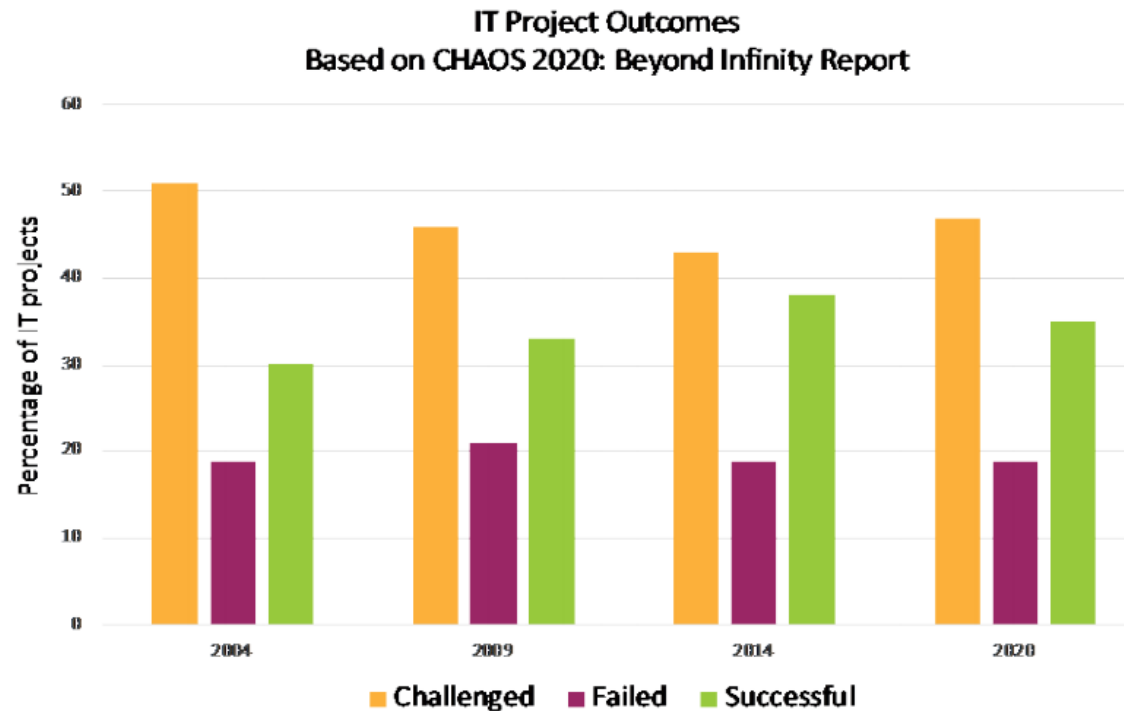
Why study software engineering?

- Software project success rate

TRADITIONAL RESOLUTION FOR ALL PROJECTS					
	2011	2012	2013	2014	2015
SUCCESSFUL	39%	37%	41%	36%	36%
CHALLENGED	39%	46%	40%	47%	45%
FAILED	22%	17%	19%	17%	19%

The Traditional resolution of all software projects from FY2011–2015 within the new CHAOS database.

*Standish Group,
CHAOS Report, 2015*



*The Cost of Poor Software
Quality in the US: A 2020
Report*

Why study software engineering?

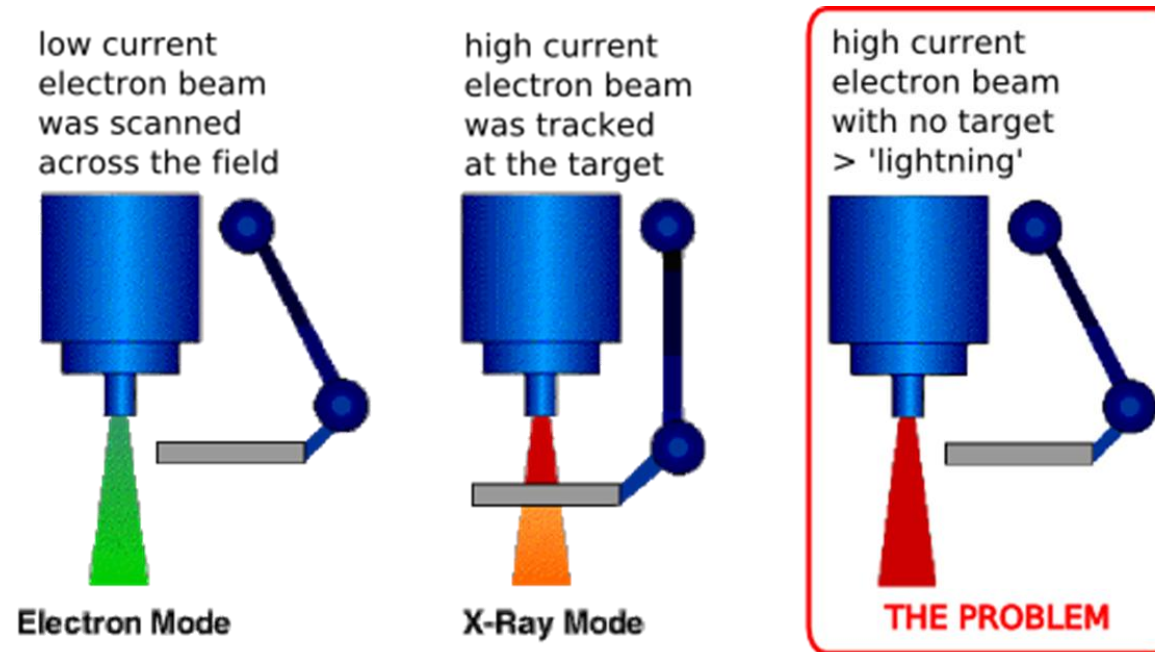
- Examples of Software Defects Leading to Failures
 - Ariane 5 Flight 501 functioned well for 40 seconds, then veered off course and was destroyed (June 4, 1996); contained four satellites: cost was \$500 million



<https://www.youtube.com/watch?v=N6PWATvLQCY>

Why study software engineering?

- Therac-25, a computer-controlled radiation therapy machine was involved in at least six accidents



tray including the target, a flattening filter, the collimator jaws and an ion chamber was moved OUT for "electron" mode, and IN for "photon" mode.

<https://www.crcpress.com/>

Why study software engineering?

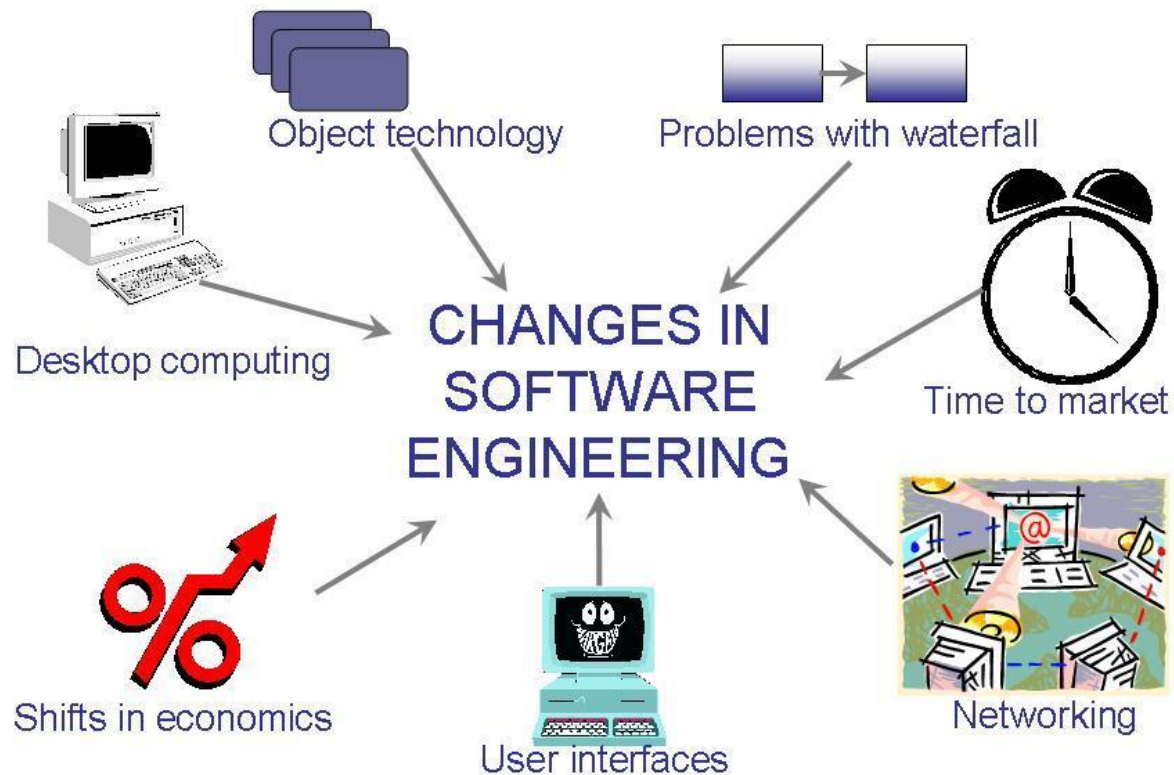
- Defect, Failure

- **Defect:** Refers to a flaw, bug, or issue in the software that may cause it to behave incorrectly. It is the result of an error or mistake during development, design, or coding.
- **Failure:** Refers to the inability of the software to perform its intended function or meet the expected requirements. A failure occurs when the software behaves incorrectly or crashes as a result of defects, external factors, or operational issues.

Why study software engineering?

- Causes of software failures
 - More complicated problems
 - Lack of appropriate training in software engineering
 - Unclear goal
 - Incomplete/wrong specification and requirements
 - Change of specification and requirements
 - Design errors, programming errors
 - Lack of planning
 - ...

Why study software engineering?



*Pfleeger and Atlee, Software Engineering:
Theory and Practice*

■ How has SE changed?

Wasserman's Seven Key Factors

1. Critically of time-to-market
2. Shifts in the economics of computing
3. Availability of powerful desktop computing
4. Extensive local- and wide-area networking
5. Availability and adoption of object-oriented technology
6. Graphical user interfaces
7. Unpredictability of the waterfall model of software development

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What is Software Engineering

■ Software

- Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
- Software includes:
 - The source code and binary/object code;
 - Documents such as requirements analysis, specification, design;
 - The procedure is used to setup and operate software system.

What is Software Engineering

Software components		Example
Program	1. Source codes 2. Object codes	
Documents	1. Analysis/Specification 2. Design 3. Implementation 4. Testing	a. Formal specification b. Context diagram c. Data flow diagram a. E-R diagram a. Code list a. Test data b. Test results
Operation procedures	1. Installation and user guide 2. Troubleshooting guide	

What is Software Engineering

■ Types of software

- *System software* is the software that acts as tools to help construct or support applications software.
 - Operating systems
 - Databases
 - Networking software
 - Compilers
 - ...
- *Applications software* is software that helps perform some directly useful or enjoyable task.
 - Games
 - Information systems
 - Real-time systems
 - Embedded systems
 - Office software
 - Scientific software
 - ...

Software can either be off-the-shelf or tailor-made for a particular application.

What is Software Engineering

- The first NATO conference (1968): **Software engineering** is the **establishment** and **use** of sound engineering principles in order to obtain economically software that is reliable and works efficiently on real machines.
- **IEEE Standard 61012-1990 Software engineering** is:
 - (1) The **application** of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software, that is, the application of engineering to software, and
 - (2) The **study** of approaches as in (1).

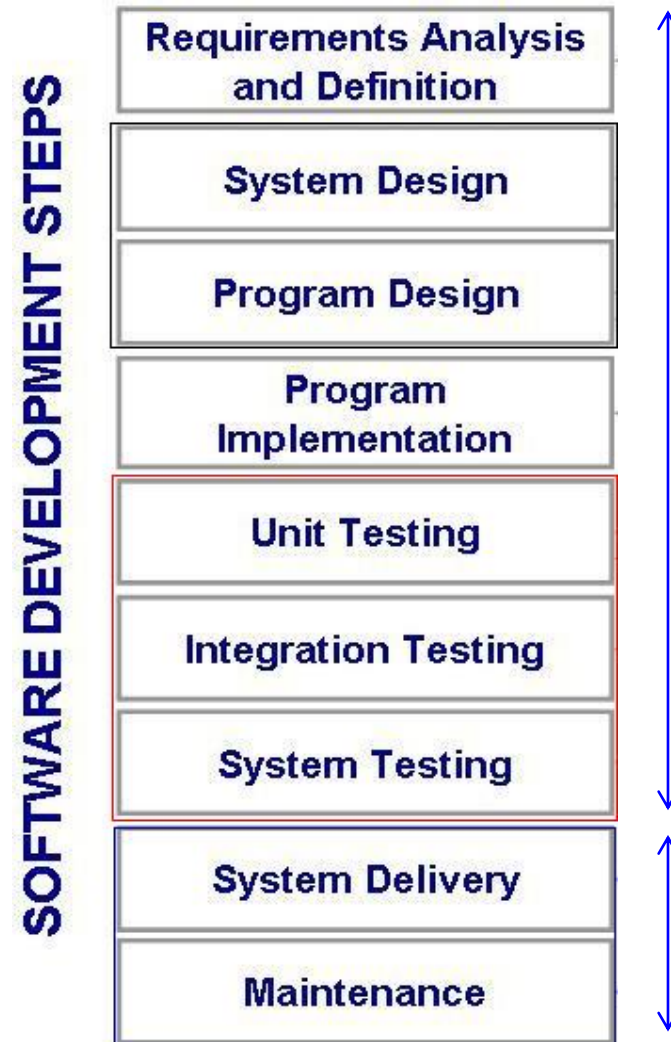
What is Software Engineering

- The goal of software engineering is to create software:
 - High quality
 - Satisfying customer requirements
 - Deploying and operating easily
 - Maintaining easily
 - On time
 - On budget
 - The price is getting lower and lower

Content

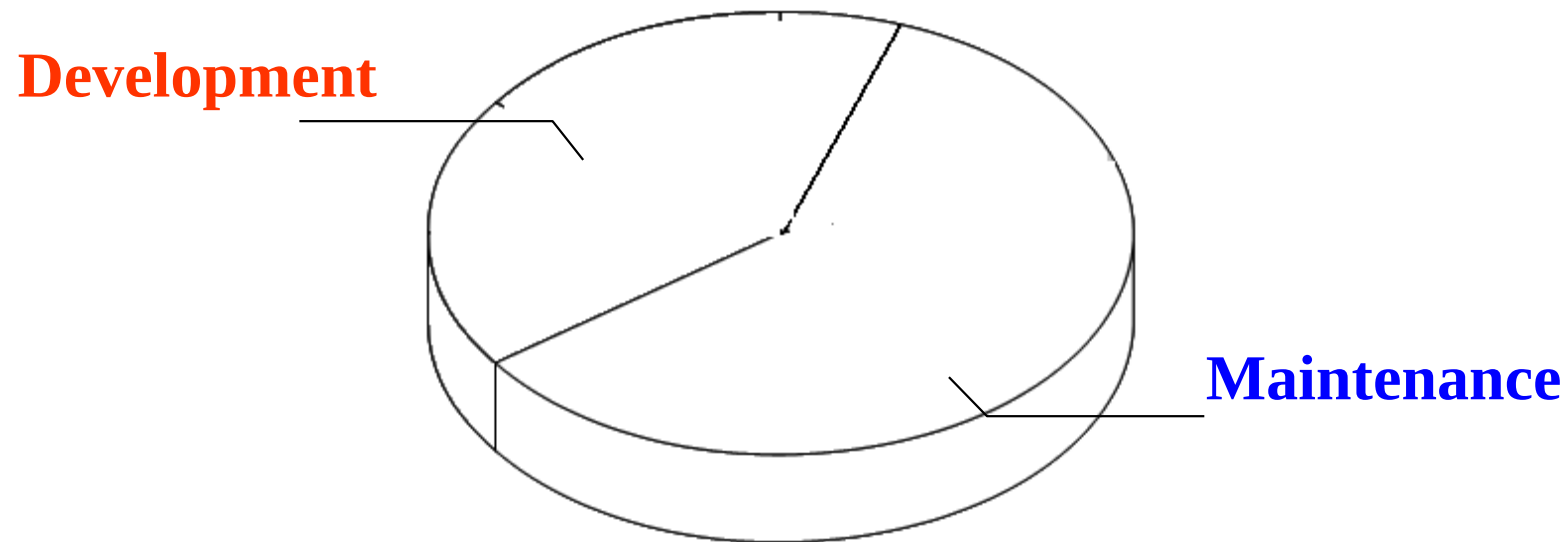
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Steps of Software Development



Steps of Software Development

- Maintenance step and various development steps
 - Maintenance alone consumes about 50-70% the total cost of a software system over its lifetime.

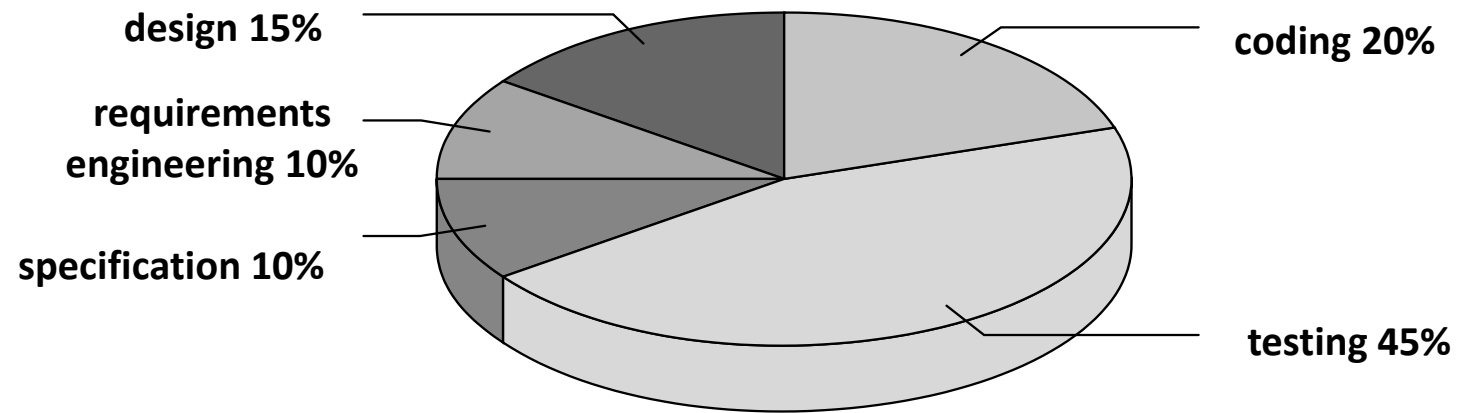


Steps of Software Development

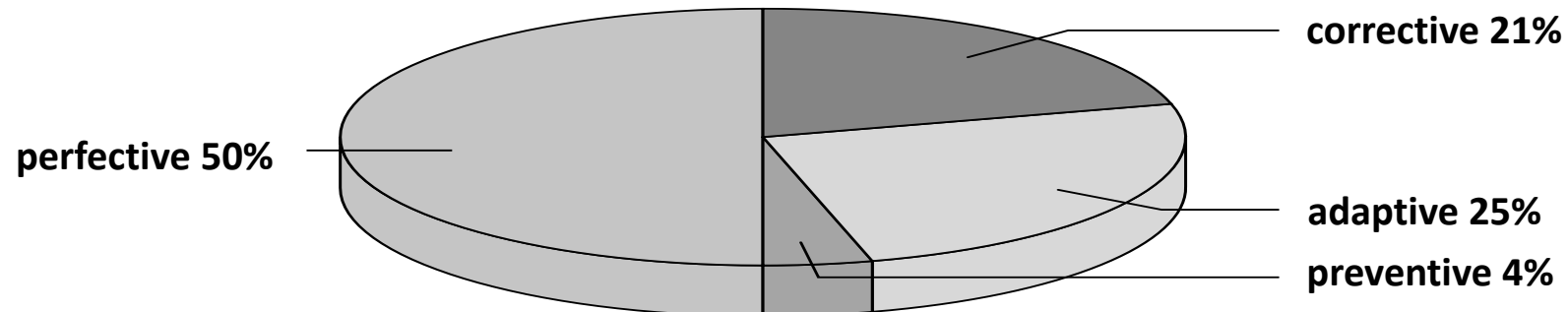
- **Requirements Analysis & Definition:** collecting customers' requirements and modeling the requirements.
- **Design:** modeling the system and detailing each module (system design and program design).
- **Implementation:** using a detailed design and choose a programming tools for coding modules.
- **Testing:** detecting errors via program testing and system testing.
- **System delivery:** deploying the system at the customer side; training and supporting clients.
- **Maintenance:** correcting errors; improving functions; modifying software to adapt to environmental change.

Steps of Software Development

- Global distribution of effort: 40 – 20 – 40

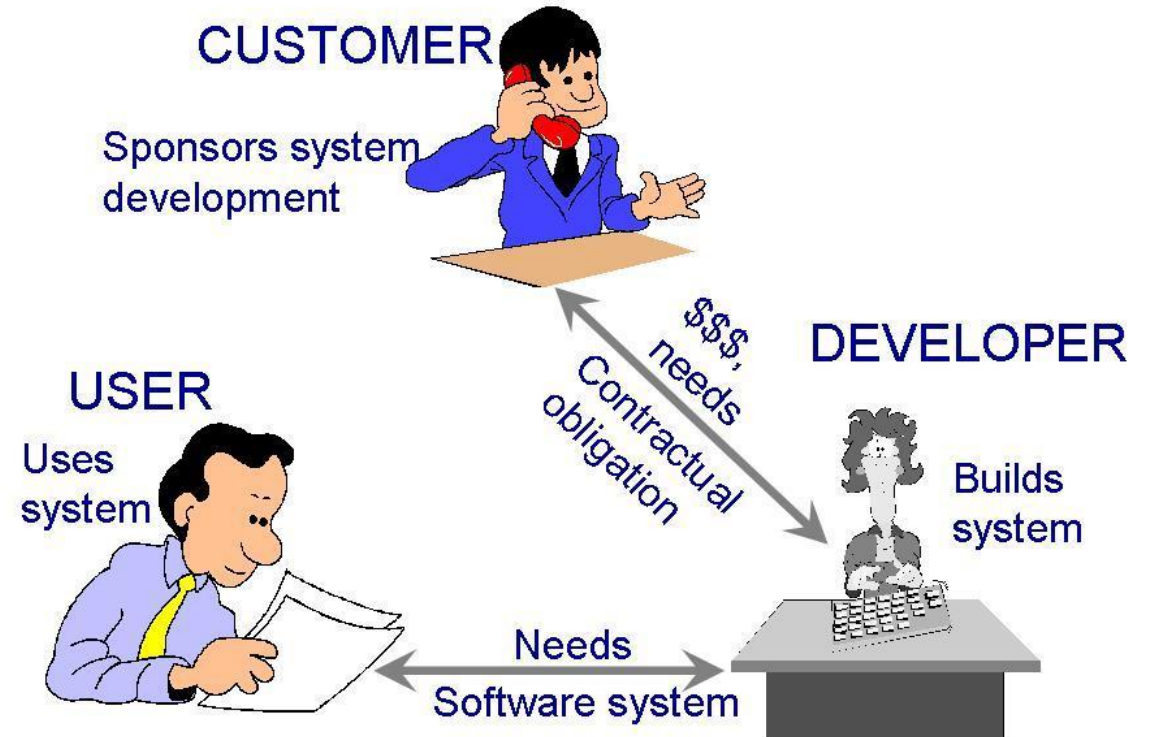


- Distribution of maintenance activities



Who Does Software Engineering?

- Customer: the company, organization, or person who pays for the software system.
 - Developer: the company, organization, or person who is building the software system.
 - User: the person or people who will actually use the system.
- **Participants** (stakeholders) in a software development project



Pfleeger and Atlee, Software Engineering: Theory and Practice

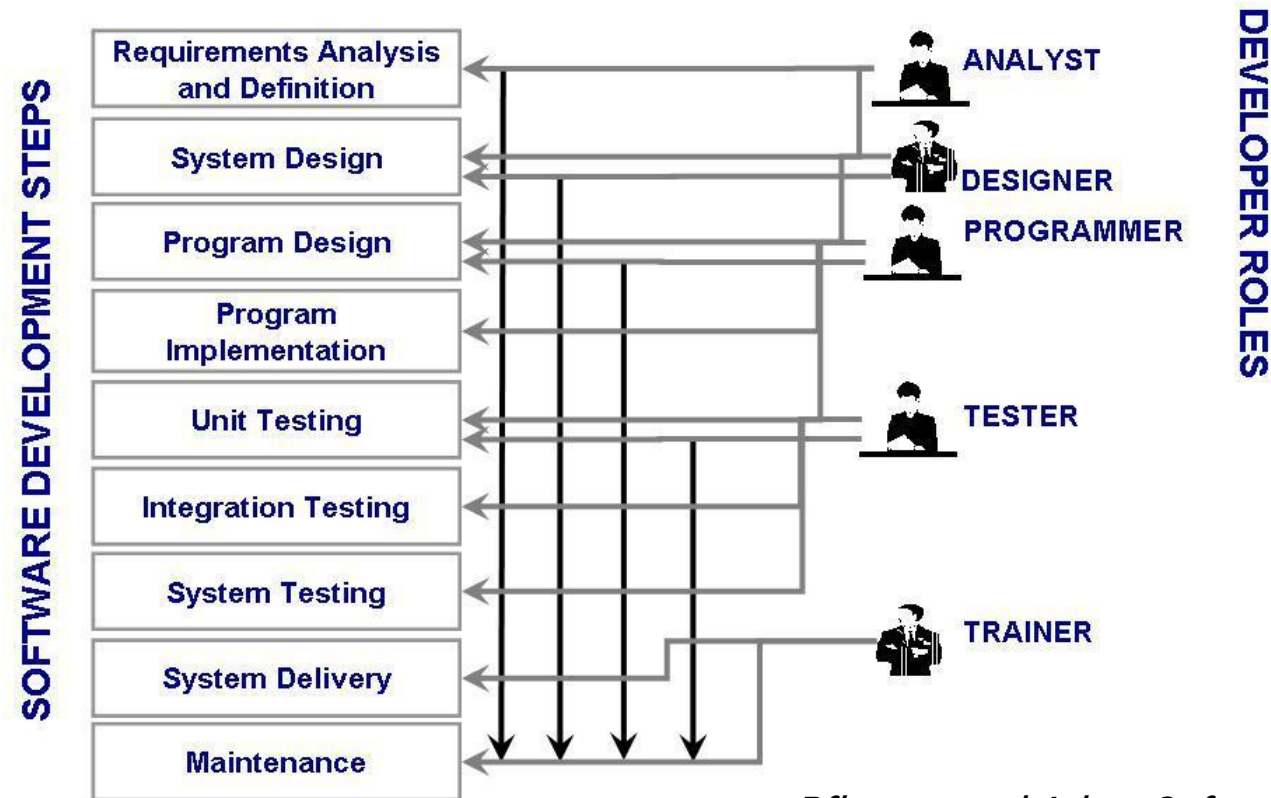
Who Does Software Engineering?

Members of the development team

- Requirement analysts: work with the customers to identify and document the requirements.
- Designers: generate a system-level description of what the system is to do.
- Programmers: write lines of code to implement the design.
- Testers: catch faults.
- Trainers: show users how to use the system.
- Maintenance team: fix faults or make changes to the system after the system has been accepted.
- Librarians: prepare and store documents such as requirements specification, design description, etc.
- Configuration management team: maintain a correspondence among various the requirement, the design, the implementation and the tests.

Who Does Software Engineering?

- Typical roles played by the members of a development team



*Pfleeger and Atlee, Software Engineering:
Theory and Practice*

Students are required to spend one week researching and writing an essay on the topic of software defects, with the following requirements:

- Content:
 - Identify 5 software defects that occurred between 2022 and 2024. For each defect, provide the name of the software and summarize the damages caused by the defect.
 - Present your thoughts, in the role of a software developer, on how to minimize the negative impact caused by these software defects.
- Formatting:
 - The essay should be written on A4 paper, with a 2cm margin on all sides (top, bottom, left, right), using the Times New Roman font, 12pt size, 1.0-line spacing, and justified text alignment.
 - The essay should consist of 2 pages:
 - Page 1: Answer the questions listed under the content section.
 - Page 2: List the references used to complete the essay.
- Submission Guidelines:
 - Naming convention for the assignment: Use the format MSSV_ LHP_ A1(.docx/.pdf), where MSSV is the student ID and LHP is the course class code.
 - Submission location: on elearning.ctu.edu.vn
 - Deadline: 5:00 PM on 11/01/2025.

Assignment 1

Q&A

References

- Shari Lawrence Pfleeger, Joanne M. Atlee, Software Engineering theory and practice, Pearson Prentice Hall, 4th edition, 2010.
- Hans Van Vliet, Software Engineering principles and practice, John Wiley, 2007.
- Hiep Xuan Huynh, Lan Phuong Phan, Introduction to Software Engineering, Can Tho university publishing house, 2010.